

Q1: Find the regular expression for the following:

- a) $L = \{w \in \{a,b\}^* \mid \text{the first symbol of } w = \text{the last symbol of } w\}$

Regular Expression: $((\epsilon + a) + a(a+b)^*a) + ((\epsilon + b) + b(a+b)^*b)$

- b) $L = \{w \in \{a,b\}^* \mid |w| \text{ is divisible by } 3\}$

Regular Expression: $((a+b)(a+b)(a+b))^*$

- c) $L = \{w \in \{a,b\}^* \mid w \text{ has alternating characters, starting with } a\}$

Regular Expression: $\epsilon + (a(ba)^*)(\epsilon + b)$

Or $(ab)^*a^* + (ab)^*$

- d) $L = \{w \in \{a,b\}^* \mid w \text{ does not contain } aa \text{ as a substring}\}$

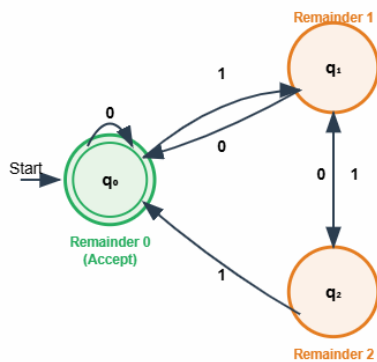
Regular Expression: $(\epsilon + a)(b + ba)^*$

- e) $L = \{w \in \{a,b\}^* \mid \text{the 3rd character of } w \text{ is } b\}$

Regular Expression: $(a+b)(a+b)b(a+b)^*$

Q2: Design a DFA over the alphabet $\Sigma=\{0,1\}$ that accepts the language:

- a) $L=\{w \in \{0,1\}^* \mid \text{the binary number represented by } w \text{ is divisible by } 3\}$



- b) $L=\{w \in \{0,1\}^* \mid \text{the substring } 0101 \text{ does not occur in } w\}$.

